*MAIDA 2027 Bridging Program*

# PYTHON PROGRAMMING ESSENTIALS

Session 3 of 5

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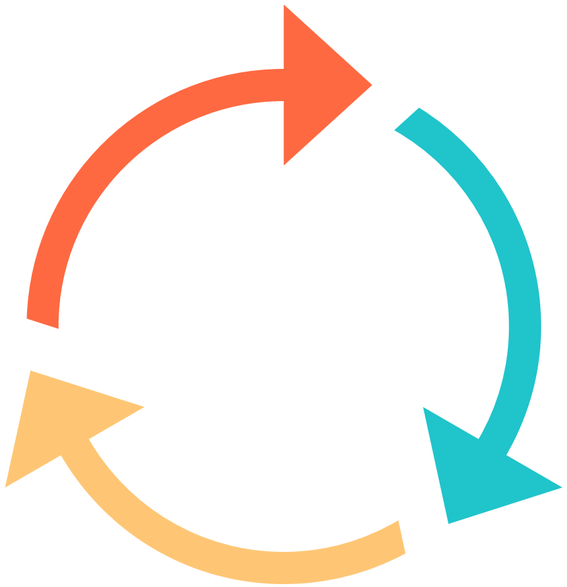
# LOOPS

In programming, a **loop** is a way to repeat a set of instructions **multiple times** until a certain condition is met (or for a fixed number of times)



# LOOPS

## The for loop

```
for [loop variable] in [iterable]:
    [        loop body        ]
```

In Python's **for** loop, we execute the **loop body** once for each item using the current value assigned to the **loop variable** as determined by the specified **iterable**.

# LOOPS

The for loop (illustrative example)

# LOOPS

## The for loop (illustrative example)

My daily morning routine

```
for day in [ MON , TUE , WED , THU , FRI , SAT , SUN ]:
```

# LOOPS

## The while loop

```
while [conditional]:
    [        loop body        ]
```

In Python's **while** loop, we execute the **loop body** as long as the specified **conditional** expression yields **True**.

# LOOPS

The `while` loop (illustrative example)

# LOOPS

## The while loop (illustrative example)

My weekly laundry routine

**while laundry basket is not empty :**

```
1 put clothes from laundry basket to washing machine
2 turn washing machine on
3 place wet clothes on clothesline
```